

* Spirals: "get rid of R " $L = c_1 R^2 I_0$

$\rightarrow$ Vir Thm

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* Ellipticals: "get rid of L "

$\rightarrow$ solve in one, plug into other

Spirals: Solve R in one $\rightarrow$ plug into other

* Sersic profiles:

$$I(r) = I_0 e^{-b \left(\left(\frac{r}{r_e} \right)^{1/n} - 1 \right)}$$

$n=1$: exponential (spirals)

$n=4$: de Vau. (ellipticals)

PSS optical:

$$\ln(R_e) \propto 1.24 \ln(\sigma) + 0.82 \ln(I_0)$$

$$\rightarrow \text{assume } \left(\frac{M}{L} \right) = \mu = L^\beta$$

$\rightarrow$ Ellipticals: $L = c_1 R_e^2 I_e$

$\leftarrow$ Plug this in

Vir Thm

$2K + U$

$$2 \left(\frac{1}{2} M_{gal} \sigma^2 \right) = c_2 \frac{G M_{gal}^2}{R}$$

$$\sigma^2 = \frac{c_2 G M_{gal}}{R}$$

From here we divide by L :

$$\frac{\sigma^2}{L} = \frac{c_2 G}{R} \left(\frac{M_{gal}}{L} \right) = \frac{c_2 G}{R} \mu$$

solve for L :

$$L = \frac{\sigma^2 R \mu}{c_2 G}$$

$$c_3 = (c_2 G)^{-1}$$

$$L = c_3 \sigma^2 R \mu$$

Given: $\mu = L^\beta$, plug in

$$L = c_3 \sigma^2 R L^\beta$$

$$L^{1-\beta} = c_3 \sigma^2 R_e$$

$$(c_2 R_e^2 I_e)^{1-\beta} = c_3 \sigma^2 R_e$$

$$c_2^{1-\beta} R_e^{2-2\beta} I_e^{1-\beta} = c_3 \sigma^2 R_e$$

$$\bullet \left(\frac{c_2^{1-\beta}}{c_3} \right) R_e^{1-2\beta} I_e^{1-\beta} = \sigma^2$$

*Take $\ln$:

$$\ln \left(\left(\frac{c_2^{1-\beta}}{c_3} \right) R_e^{1-2\beta} I_e^{1-\beta} \right) = \ln(\sigma^2)$$

$$\ln \left(\frac{c_2^{1-\beta}}{c_3} \right) + \ln(R_e^{1-2\beta}) + \ln(I_e^{1-\beta}) = \ln(\sigma^2)$$

Some Constant

C

$$C + 1-2\beta \ln(R_e) + 1-\beta \ln(I_e^{1-\beta}) = 2 \ln(\sigma)$$

$$\ln(R_e) = \left(\frac{2}{1-2\beta} \right) \ln(\sigma) - \frac{1-\beta}{1-2\beta} \ln(I_e) + \hat{C}$$

$$\hat{C} = C/(1-2\beta)$$

$$\ln(R_e) = \underbrace{\left(\frac{2}{1-2\beta} \right)}_{1.24} \ln(\sigma) - \underbrace{\left(\frac{1-\beta}{1-2\beta} \right)}_{0.82} \ln(I_e) + \hat{C}$$

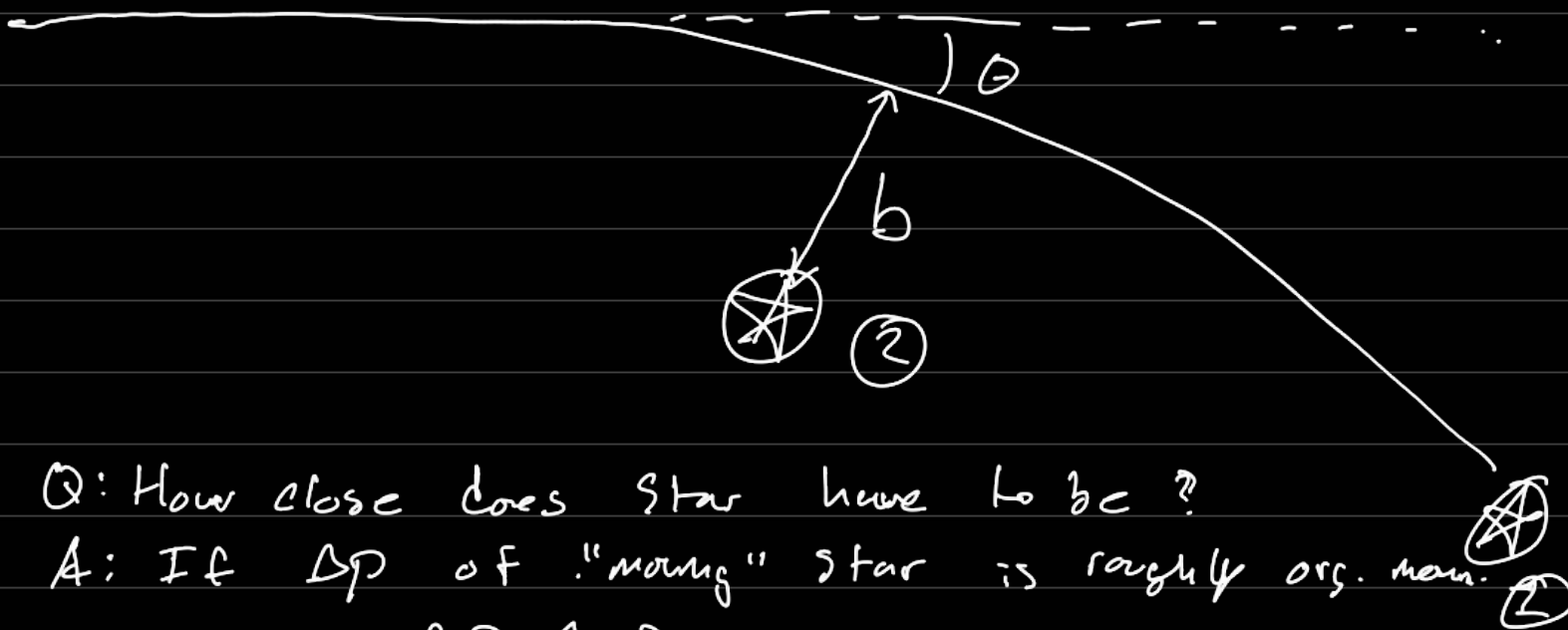
1.24

0.82

* Calculator:

Gravitational Coll:

• Not "physical" coll: think slug-slug



Q: How close does star have to be?

A: If Δp of "moving" star is roughly org. mom. (2)

* "Deep" analysis: we'll do a "simple"

$$\Delta p = p_i = m_i v_i$$

• Grav. Force: $F_{21} = \frac{\Delta p}{\Delta t} = \frac{m_i v_i}{\Delta t}$

$$F_{12} = \frac{G m_1 m_2}{b^2}$$

• Assumption: masses are similar:

$$F_{12} = \frac{G m^2}{b^2} = \frac{m_i v_i}{\Delta t}$$

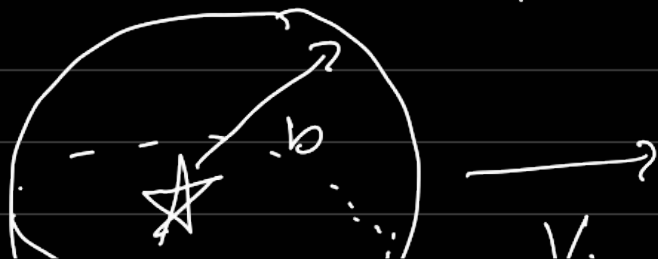
• "Impact" time: $b = v \Delta t$

$\Delta t = b/v$: plug into Force

$$\frac{G m^2}{b^2} = \frac{m_i v_i}{b/v} = \frac{m v^2}{b} \Rightarrow \frac{G m}{b} = v^2$$

$$b = \frac{G m}{v^2}$$

* Assume that m_i :



$$X = V \Delta t$$

→ Star sweeps out a Volume / per time

$$\frac{(\text{Vol})}{\Delta t} = \frac{4\pi b^2 X}{\Delta t} = \frac{4\pi b^2 V \Delta t}{\Delta t}$$

$$\frac{(\text{Vol})}{\Delta t} = 4\pi b^2 V_1$$

→ Astro: $n = \frac{N}{(\text{Vol})}$ (# - densities)

* Define: (# of coll / time) = $\frac{N}{\Delta t}$

$$\frac{N}{\Delta t} = n \left(\frac{\text{Vol}}{\Delta t} \right) = n 4\pi b^2 V_1$$

$$* \frac{N}{\Delta t} = n 4\pi b^2 V_1 \quad \left(\frac{\text{\# of coll}}{\text{Time}} \right)$$

• $N=1$: When a single collision occurs

↳ $\Delta t \equiv \tau_{\text{rel}}$

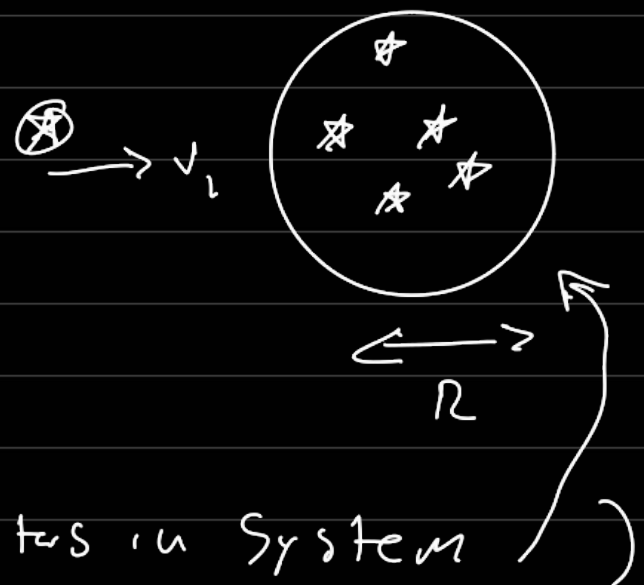
$$\tau_{\text{rel}} = \frac{1}{n 4\pi b^2 V_1}$$

* Idea: time to cross

Q: time to cross

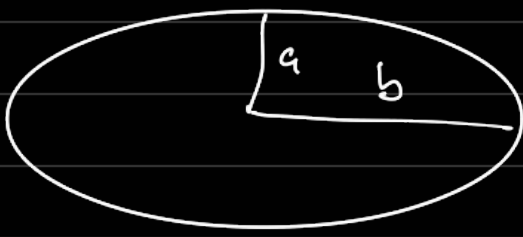
$$\tau_{\text{cross}} = R/V$$

$$\frac{\tau_{\text{cross}}}{\tau_{\text{rel}}}$$



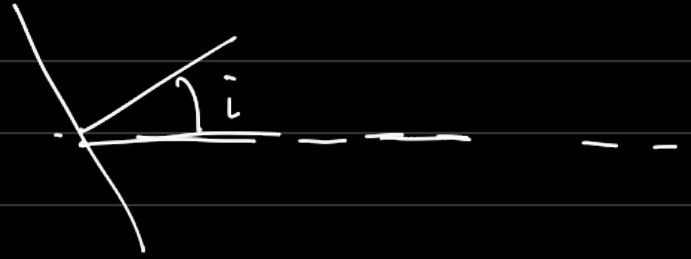
$$* \frac{\tau_{\text{rel}}}{\tau_{\text{cross}}} \propto N \quad (\text{\# of stars in system})$$

* Might not have mentioned:



• Inc. angle: i

$$\sin(i) = b/a$$



* Give

$\rho(r)$:

$$M(r) = \int_0^r 4\pi r^2 \rho(r) dr$$

